

TheoryAssignment_2 Solutions

1. What is CSS Box Model?

The **CSS Box Model** is a container that wraps around every HTML element. It consists of four main components from innermost to outermost:

- **Content:** The actual content of the box, such as text or images.
- **Padding:** Clears an area around the content; it is transparent.
- **Border:** A border that goes around the padding and content.
- **Margin:** Clears an area outside the border; it is transparent and used to create space between elements.

2. What is media query? Explain with a sample code.

A **media query** is a CSS technique introduced in CSS3 that allows content to adapt to different screen resolutions, pixel ratios, and device types (like phones, tablets, and desktops).

CSS

```
/* Base styles for all devices */
```

```
body {
```

```
    background-color: lightblue;
```

```
}
```

```
/* Styles for screens that are 600px wide or less */
```

```
@media screen and (max-width: 600px) {
```

```
    body {
```

```
        background-color: lightyellow;
```

```
    }
```

```
}
```

3. What are the features of Bootstrap?

- **Responsive Design:** Provides fluid grid systems that appropriately scale up to 12 columns as the device or viewport size increases.
- **Mobile-First Approach:** Includes mobile-first styles throughout the core library by default.
- **Pre-built Components:** Offers extensive pre-styled components like buttons, navbars, modals, and dropdowns.
- **JavaScript Plugins:** Comes with built-in jQuery/vanilla JavaScript plugins for interactive components.

- **Customizability:** Easily customizable via Sass variables and utility classes.

4. Write an example of basic grid structure in Bootstrap?

Bootstrap's grid system uses a series of containers, rows, and columns to layout and align content.

HTML

```
<div class="container">

  <div class="row">

    <div class="col-sm-4 bg-primary text-white">Column 1 (Width 4)</div>

    <div class="col-sm-4 bg-secondary text-white">Column 2 (Width 4)</div>

    <div class="col-sm-4 bg-success text-white">Column 3 (Width 4)</div>

  </div>

</div>
```

5. What are CSS Selectors?

CSS Selectors are patterns used to select and target HTML elements on a web page so that styles can be applied to them. Common types include:

- **Element/Tag Selector:** Selects elements based on the tag name (e.g., `p { color: red; }`).
- **ID Selector:** Selects an element based on its unique id attribute using `#` (e.g., `#header { background: blue; }`).
- **Class Selector:** Selects elements with a specific class attribute using `.` (e.g., `.btn { padding: 10px; }`).
- **Universal Selector:** Selects all elements using `*`.

6. What are CSS backgrounds, list the properties?

CSS background properties are used to add background effects to elements. The primary properties are:

- **background-color:** Sets the background color of an element.
- **background-image:** Specifies one or more background images for an element.
- **background-repeat:** Defines how background images are repeated (e.g., `repeat`, `no-repeat`).
- **background-attachment:** Sets whether a background image scrolls with the rest of the page or is fixed.
- **background-position:** Specifies the starting position of a background image.
- **background-size:** Specifies the size of the background image elements.

7. How can we overlap elements in CSS?

Elements can be overlapped in CSS using the **position** property combined with the **z-index** property.

- By setting an element's position to absolute, relative, or fixed, you can use top, left, right, or bottom to place it over another element.
- The z-index property controls the stacking order (which element appears on top of the other). Higher numbers sit on top of lower numbers.

8. What are the different positioning properties in CSS?

The position property specifies the type of positioning method used for an element. The five different values are:

- **static:** Default positioning; elements render in the normal document flow.
- **relative:** Positioned relative to its normal position without altering surrounding layout flow.
- **absolute:** Positioned relative to the nearest positioned ancestor (instead of positioned relative to the viewport).
- **fixed:** Positioned relative to the browser window, meaning it stays in the same place even if the page is scrolled.
- **sticky:** Positioned based on the user's scroll position, acting like relative until a given offset point is met, then acting like fixed.

9. What is ECMAScript?

ECMAScript is a scripting language specification standardized by Ecma International. It was created to standardize JavaScript, ensuring that different browsers and environments support the same core language features. Modern versions like ES6 (ECMAScript 2015) introduced major features like arrow functions, classes, promises, and let/const declarations.

10. What is the difference between var, let and const in JavaScript. Explain Temporal Dead zone?

Feature	var	let	const
Scope	Function scope	Block scope	Block scope
Hoisting	Yes (initialized with undefined)	Yes (not initialized)	Yes (not initialized)
Re-declaration	Allowed	Not allowed	Not allowed
Re-assignment	Allowed	Allowed	Not allowed

Temporal Dead Zone (TDZ): The Temporal Dead Zone is a behavior that occurs with variables declared using let and const. It is the time period between the entering of a scope (block) and the actual variable declaration being evaluated/initialized. During this window, attempting to access the variable results in a ReferenceError